

INTERNSHIP REPORT
ON
JOB PORTAL RECOMMENDATION SYSTEM

Submitted to

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

**BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND
ENGINEERING**



By

ROLL NO: CS-23411036
NAME: ANJALI TIWARI

Batch: 2026-2027

CANDIDATE'S DECLARATION

I, Anjali Tiwari, do hereby declare that the internship report titled “Job Portal Recommendation System” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references used in the preparation of this report have been duly quoted and acknowledged, and I have not taken as such any material from any other source without proper citation.

I affirm that this report is the result of my own work carried out during my internship and has not been submitted, in part or in full, to any other institute or university for the award of any other degree or diploma. I shall be solely responsible for any kind of copyright violation, if found, in this regard.

Signature :

Student Name: Anjali Tiwari

Roll No.: CS-23411036

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to everyone who supported and guided me throughout my internship at DigiSamaksh Pvt. Ltd. This internship provided me with a valuable opportunity to apply my academic knowledge to a practical software development project and to understand the professional practices followed in the technology industry.

I am thankful to the management and technical team of DigiSamaksh Pvt. Ltd. for giving me the opportunity to work on the Job Portal Recommendation System. Their guidance, constructive feedback and technical discussions helped me understand the different stages of software development, including requirement analysis, system design, backend development, database integration, AI-based functionality and testing.

I would also like to express my sincere thanks to my faculty members and the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous academic guidance and encouragement throughout the internship period. I am also grateful to my parents and friends for their constant support and motivation.

Finally, I am thankful to everyone who, directly or indirectly, contributed to the successful completion of my internship and this report.

Signature

Student Name: Anjali Tiwari

Roll No.: CS-23411036

INTERNSHIP COMPLETION CERTIFICATE

Employee ID: DIGI/DI/2026/0168

Reference No.: DIGI/IOL/0908

INTERNSHIP COMPLETION CERTIFICATE TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Anjali Tiwari**, s/d/o **Mr. Surya Prakash Tiwari**, has successfully completed an internship as **Software Engineering Intern** in the **Technology & Development Department** at **DIGISAMAKSH Private Limited**. The internship tenure was from June 01, 2026 to August 15, 2026.

During the internship period, **Anjali Tiwari** has shown a high level of commitment, technical understanding, and problem-solving ability while working on assigned projects and tasks.

Key Responsibilities & Contributions

- ✦ **Software Development Support:** Assisted in coding, debugging, and testing software applications while ensuring adherence to project guidelines.
- ✦ **Technical Documentation:** Prepared and maintained project-related documentation including code comments, user guides, and technical reports.
- ✦ **Research & Learning:** Conducted research on new technologies, frameworks, and tools to support the ongoing development work of the department.
- ✦ **System Maintenance:** Supported the team in troubleshooting issues, performing system updates, and ensuring smooth functioning of technical infrastructure.
- ✦ **Team Collaboration:** Worked closely with developers, testers, and project managers to complete assigned modules and contribute effectively to project milestones.
- ✦ **Innovation & Problem-Solving:** Suggested improvements in workflows and participated in brainstorming sessions to enhance project outcomes.

Internship Evaluation

Based on the overall performance, learning attitude, and contributions during the internship, **Anjali Tiwari** has been evaluated on the following parameters:

- ✦ **Technical Skills:** Demonstrated good knowledge of IT concepts, coding practices, and problem-solving.
- ✦ **Work Quality:** Delivered assigned tasks with accuracy and within timelines.
- ✦ **Communication Skills:** Maintained professional communication and clarity while coordinating with team members.
- ✦ **Learning & Adaptability:** Showed eagerness to learn new skills and adapt to evolving project requirements.
- ✦ **Professional Conduct:** Maintained discipline, punctuality, and a positive approach throughout the internship.

Overall, **Anjali Tiwari** has successfully met the expectations of the internship program and has added value to the Technology & Development Department of DIGISAMAKSH Private Limited. We wish **Anjali Tiwari** success in all future endeavors.

For

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CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

The process of searching for suitable jobs has traditionally relied on generic job portals that display listings using simple keyword search, often leaving candidates to manually sift through hundreds of irrelevant postings to find the ones that truly match their skills and interests. During my internship at DigiSamaksh Pvt. Ltd., I worked on the design and development of a Job Portal Recommendation System, a web application that helps job seekers discover relevant job opportunities through a personalised, intelligent recommendation engine.

The system was built to serve two primary categories of users: job seekers and recruiters/companies. It centralizes candidate profile management, job postings, skill-based filtering, application tracking, while a recommendation layer analyses a candidate's skills, experience and preferences to suggest the most relevant job listings. The objective of the internship was to gain hands-on exposure to full-stack web development and to understand how recommendation algorithms and machine learning techniques can be practically integrated into a real-world application.

4.2 Organization Profile

DigiSamaksh Pvt. Ltd. is a technology-driven organization engaged in the development of custom software solutions, web and mobile applications, and data-driven products for clients across education, enterprise and public-sector domains. The organization focuses on building scalable digital platforms using modern technology stacks and follows an agile, collaborative approach to software development.

The internship was carried out under the guidance of the technical team of DigiSamaksh Pvt. Ltd., which provided mentorship in software engineering practices such as requirement gathering, modular system design, version-controlled development, code reviews and iterative testing. The team's exposure to real client-facing projects gave practical insight into how academic concepts of software engineering and machine learning are applied in a professional development environment.

4.3 Problem Statement

Most conventional job portals rely on basic keyword search and manual filters, which forces candidates to browse through a large number of unrelated listings before finding a suitable opportunity. This leads to several recurring problems: irrelevant job recommendations that do not match a candidate's actual skill set, increased time spent searching for the right opportunity, difficulty in discovering openings aligned with a candidate's experience and interests, lack of a centralized profile that captures a candidate's skills and preferences, and limited ability of recruiters to reach candidates who are the best fit for a given role.

The project therefore addressed the need for a unified, intelligent platform that could analyse a candidate's profile and use recommendation techniques to surface the most relevant job postings, reducing search effort for job seekers and helping recruiters connect with better-matched candidates.

4.4 Project Objectives

- To design and develop a centralized web-based job portal for job seekers and recruiters.
- To provide role-based access for job seekers and recruiters/companies.
- To capture and analyse candidate profile data such as skills, experience and preferences to build accurate recommendation inputs.
- To integrate a recommendation engine based on content-based and collaborative filtering techniques for matching candidates with relevant job postings.
- To enable online job application submission, tracking and status updates for candidates.
- To generate reports and analytics on job postings and application trends for recruiters.
- To ensure a secure, scalable and user-friendly system using modern web technologies.

4.5 Scope of the Project

The scope of the system covers the complete job search and recommendation lifecycle, starting from candidate registration and profile creation, through job posting by recruiters, skill-based filtering, personalised recommendation generation, and application submission and tracking. The system is designed to be scalable so that it can be extended to accommodate multiple job categories, industries and locations.

The current scope focuses on core job portal functionalities and the recommendation module; features such as native mobile applications, chatbot-based career guidance and advanced analytics dashboards were identified as potential areas for future enhancement beyond the internship period.

4.6 Technologies and Tools Used

The following technologies and tools were used during the development of the project:

- Frontend: React.js, HTML5, CSS3, JavaScript, Bootstrap
- Backend: Python (Django/Flask) and Node.js (Express.js) for REST API development
- Database: MySQL for relational data storage
- AI/Machine Learning: Python, Scikit-learn, Pandas, NumPy, TF-IDF and cosine-similarity based content filtering, along with collaborative filtering techniques for job recommendation
- Version Control: Git and GitHub for source code management and collaboration
- Development Tools: Visual Studio Code, Postman (API testing), XAMPP/MySQL Workbench
- Deployment: Cloud-based hosting for staging and testing environments

4.7 System Architecture

The system follows a three-tier client-server architecture consisting of a presentation layer, an application/business logic layer and a data layer. The presentation layer is a responsive web interface used by job seekers and recruiters. The application layer exposes RESTful APIs that handle authentication, profile management, application workflows and communicate with the recommendation module for job matching and ranking. The data layer uses a relational database to store candidate profiles, job postings, applications and recommendation scores.

The recommendation module operates as a separate service that receives candidate profile data and job posting data, extracts relevant features such as skills, experience and preferences, and computes a similarity/matching score used to rank and recommend the most suitable job postings to a given candidate. This modular design allows the recommendation component to be updated or retrained independently of the core job portal application.

4.8 Methodology

The project was developed following an iterative and incremental (agile) methodology, broken down into the following phases:

- Requirement Analysis: Studying existing job portals and gathering functional requirements for job seekers and recruiters.
- System Design: Designing the database schema, API endpoints, module structure and user-interface wireframes.
- Frontend Development: Building responsive interfaces for registration, dashboards, job listings, recommendations and application tracking.
- Backend Development: Implementing authentication, role-based access control, job posting/application logic and REST APIs.
- Recommendation Module Development: Building a candidate-job matching model using content-based and collaborative filtering techniques.
- Integration: Connecting the frontend, backend and recommendation module into a single working system.
- Testing: Performing unit testing, integration testing and user-acceptance testing to validate functionality.
- Deployment and Review: Deploying the application on a staging environment and reviewing it with the mentoring team.

4.9 Expected Outcomes

- A fully functional web-based job portal with separate portals for job seekers and recruiters.

- A personalised recommendation engine that reduces the manual effort of browsing irrelevant job listings.
- A recommendation module capable of ranking job postings by relevance to a candidate's profile.
- Reduced time spent by candidates in searching for suitable jobs and improved relevance of job applications.
- A centralized, reliable repository of candidate and job posting data that can support future reporting and analytics needs.
- Practical, hands-on experience in full-stack development and applied machine learning within a professional software development environment.

4.10 Certificates of Completion and Other Communication Mail Proof

The Internship Completion Certificate issued by DigiSamaksh Private Limited, certifying the successful completion of the internship by Anjali Tiwari as a Software Engineering Intern in the Technology & Development Department (tenure: June 01, 2026 to August 15, 2026), is attached in Chapter 3 of this report. The Offer Letter issued by DigiSamaksh Private Limited confirming the offer of internship is attached below.

**OFFER LETTER
TO WHOMSOEVER IT MAY CONCERN**

☎: 91+(0)9582349641; [in/company/digisamaksh](https://www.linkedin.com/company/digisamaksh/); [ig/digisamaksh](https://www.instagram.com/digisamaksh/) [fb/digisamaksh](https://www.facebook.com/digisamaksh/)

CHAPTER 5

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